DATA MINING TASKS

Concept Map Activity

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Topic: Classification of Data Mining Tasks into Predictive and Descriptive Tasks

1. Introduction

Data mining is the process of discovering useful patterns from large collections of data. The work it performs is called a data mining task, and these tasks are divided into two broad groups. Predictive tasks use the data we already have to make a statement about values we do not yet know. Descriptive tasks summarise the data to expose the patterns that characterise it.

2. Task Descriptions and Examples

Task	Category	Brief Description	Real-World Example
Classification	Predictive	Learns from records whose class is already known and then assigns a new record to one of those classes.	An email system marking each incoming mail as spam or not spam.
Prediction	Predictive	Builds a model that estimates a continuous number for an unseen record from its input values.	A telecom company estimating next month's bill of a customer from past usage.
Clustering	Descriptive	Groups objects so that members of a group are alike and different from other groups, without any predefined labels.	A bank grouping its customers into saver, spender and investor segments.
Association Rule Mining	Descriptive	Finds if-then relationships between items that frequently occur together in transactions.	An online store showing 'customers who bought a phone also bought a cover'.

Task	Category	Brief Description	Real-World Example
Outlier Analysis	Descriptive	Detects objects that behave very differently from the rest of the data; useful for finding errors and fraud.	An insurance company catching an unusually large, suspicious claim.
Evolution Analysis	Descriptive	Studies data whose behaviour changes with time to describe trends, cycles and deviations.	A weather department studying the rainfall pattern of a city over 50 years.

3. Relationships Between the Tasks

The links drawn in the concept map are explained below.

Link	Meaning
Data Mining Tasks → Predictive / Descriptive	Every task is placed under one of the two groups: predictive tasks forecast unknown values, descriptive tasks summarise existing data.
Predictive → Classification and Prediction	Both learn from past data; classification outputs a category, prediction outputs a number.
Descriptive → Clustering, Association, Outlier, Evolution	These reveal hidden structure in data without a target value to forecast.
Clustering → Outlier Analysis	Objects lying far from every cluster are treated as outliers.
Clustering → Classification	Clusters discovered can later serve as class labels for a classifier.
Evolution Analysis → Prediction	Trends found over time are extended to forecast future values.

4. Short Answers

(a) Predictive tasks

Predictive tasks are supervised in nature: the model is trained on past records that already carry a known result. Classification answers a 'which group?' question, while prediction answers a 'how much?' question.

(b) Descriptive tasks

Descriptive tasks work mostly without a known result. They tell us what is going on inside the data: which records are alike (clustering), which items travel together (association), which records are unusual (outlier analysis) and how behaviour shifts over time (evolution analysis).

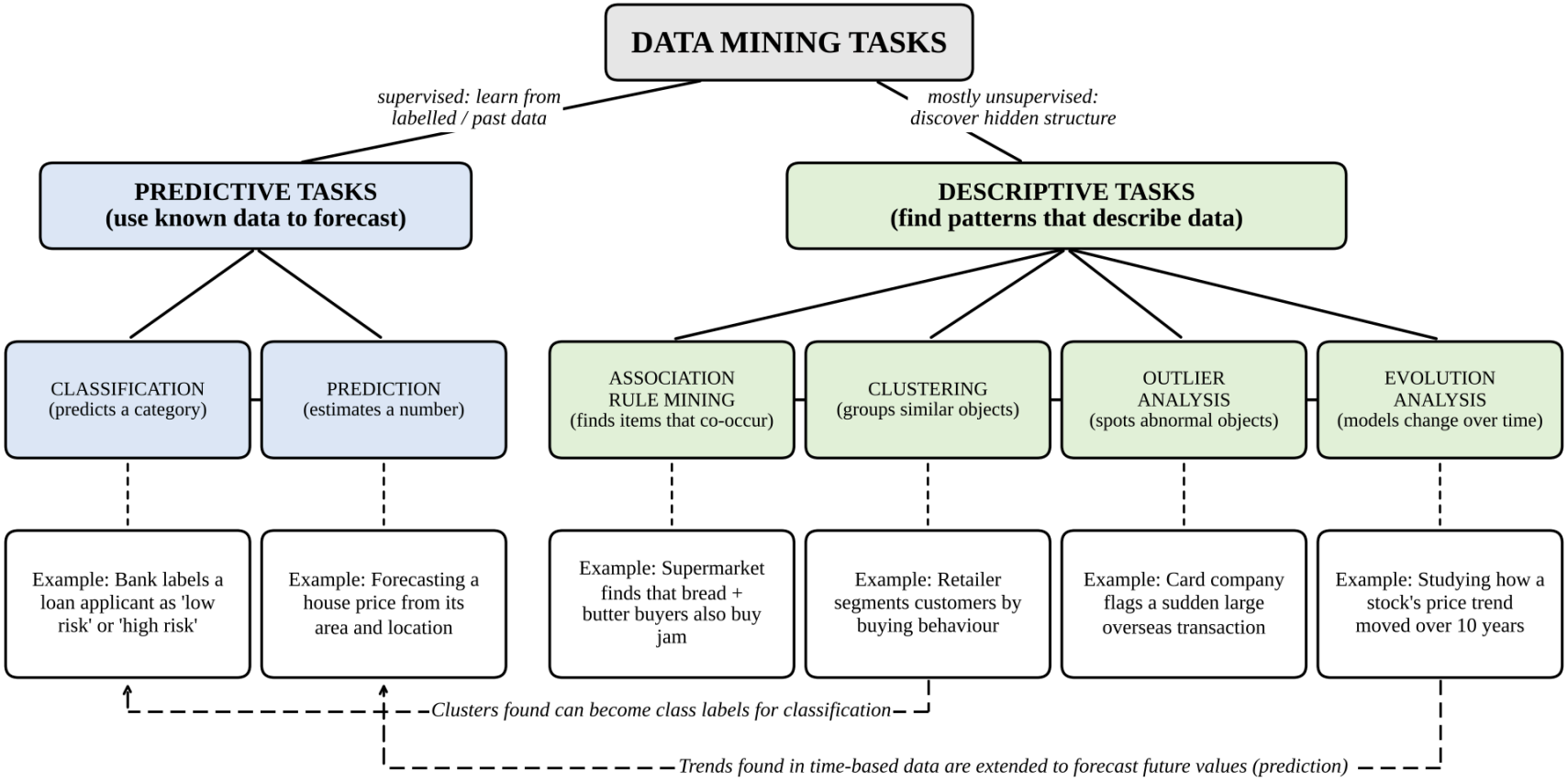
(c) Why the distinction matters

Choosing between the two groups depends on the question being asked. If the business needs a forecast, a predictive task is selected; if it needs insight into existing behaviour, a descriptive task is selected.

5. Conclusion

Data mining tasks can be organised as a tree with two main branches, predictive and descriptive. The six tasks differ in purpose, yet they support one another: clusters feed classifiers, outliers are measured against clusters, and trends from evolution analysis help prediction. The concept map on the next page shows this hierarchy together with an example for every task.

Concept Map: Data Mining Tasks



Solid line = hierarchy / direct link Dashed line = example or cross-task relationship